



Out of Time

The year 2005 saw the clock run out on clockspeed. With *faster* leading to *hotter*, speed was no longer the rallying call for processors. Here, we explore the nature of the wall the processors hit, and the solutions which promise to bring that wall down

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R.I.P. Moore's Law
Faithful servant to a multi-billion dollar industry
1965-2005

Ok, so the above is a bit of an exaggeration. Yeah, fine, slightly more than an overstatement. But everyone is allowed a bit of hyperbole every now and then, and the events of the past few years have seen the computer processor industry do such a stark turning-on-a-dime, that only hyperbole can help juxtapose where we are today, in relation to where we were supposed to be!

10-GHz processors were supposed to be the destination, not too long ago. The number is so mind-numbing-

ly huge that when Intel revealed its plans for going faster and faster, everyone listened and cheered them on. 10 GHz was supposed to be the legacy of the Netburst architecture—the foundation of the now-doomed Pentium 4 project. Everything went fine-ish up to the 3 GHz mark and then Netburst fell flat. The Prescott, Netburst's final hurrah, made it to the 3.8-GHz mark, wheezing and numerous revisions later. 3.8 GHz was a power nightmare though—remember those videos of frying eggs on the processor—the processor lost as much as 115 watt of power! Something was terribly wrong with where we were heading.

Nature Of The Beast

The microprocessor is a complex beast. Nothing is as charming and straightforward inside the belly of this

beast as Moore's Law would have you believe: "The complexity for minimum component costs has increased at a rate of roughly a factor of two per year..." which leads to the number of transistors on a chip will double about every two years. Of course, Gordon Moore never proposed it as a simple mantra to drive the semiconductor industry forward. Moore's law was perhaps a simple roadmap for the industry itself, and an ideal PR line—clever and simple enough to keep the media happy and intrigued. Hidden behind the popular headlines of "faster every year!" though, were a plethora of engineering hurdles barely cleared, or threatening to rear their ugly heads. And much like the mythical Hydra—these nanometre beasts would sprout another head, as soon as you chopped down one problem.

Sooner or later the difficulties in going faster would catch up with the industry, and with Netburst, they did. So what were these issues?

The biggest problem, briefly hinted by the example of Prescott above—is power dissipation. Fast chips get hot, but the problem is not that simple. There are two main elements that constitute a microprocessor—capacitors and transistors, connected together through myriad wires. We are once again oversimplifying things here, but for our understanding of the issues, this simplification serves us well.

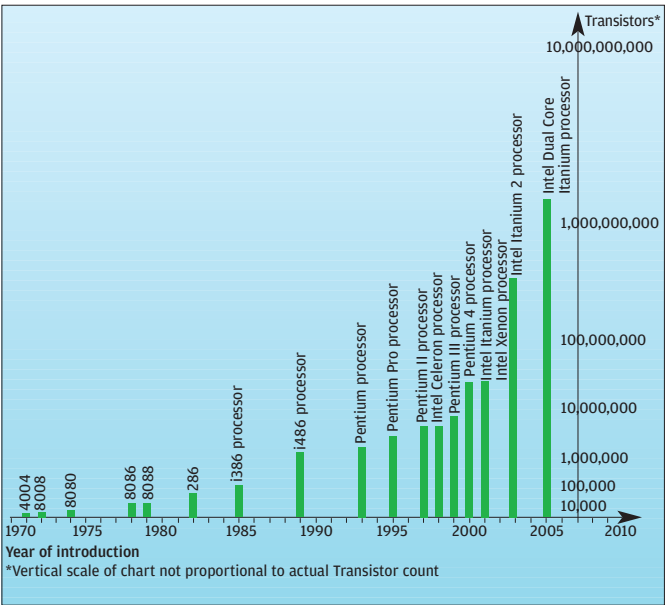
The power dissipated by a processor increases linearly with the effective capacitance, the frequency at which the processor is running and also on how much load it is carrying. The more 'processor-intensive' a program you run, the hotter the processor will get, pretty logical, right? While the last factor is pretty self-explanatory, the processor's relationship with its constituent transistors, wires and capacitors is slightly more complex.

A processor's power dissipation also increases quadruple with the core voltage of the processor. The relation can be put forth as follows:

$$\text{Power} \sim [\text{Capacitance}] [\text{Voltage}]^2 [\text{Load}] [\text{Frequency}]$$

As technology progresses, processors shrink in size—this is termed a die shrink, the die being the physical medium on which the processor's constituent elements reside and run. Shrinking dies have an interesting effect on the power dissipation of a processor. As processors get smaller and smaller, you need less voltage to run the parts. But with this shrinking die, a processor also adds features, their capabilities increase—this is progress after all. This leads to an increase in the number of transistors that are packed into the shrinking die. The increase in the transistors (capacitance), more often than not, offsets any power-reduction that a decrease in core voltage brings to the table.

One of the techniques used by engineers is called clock gating. To alleviate the power problems caused by an increase in transistors, engineers design the system such that parts of the processor are only activated when the system requires work off them. Otherwise, they hang



Rising transistor counts and underlying processor manufacturing problems put the brakes on faster processors

idle, thus reducing the power draw. Techniques like these are used to somewhat reign in the runaway power draw. Generally speaking, power drawn is made to increase linearly, while the number of transistors used increases exponentially.

Power drawn is thus a controllable beast. Plus, it is actually used to do real work by the processor. The real issue lies with power leakage. While power drawn is linear and controllable, as the die shrinks, the leakage component increases quickly and exponentially, threatening to overwhelm and destroy the circuitry if not reigned in. As the core voltage increases and the size of the processor die reduces, the bigger the problem of leakage gets. Power leaks away as current. As the current leaks, the system has to draw more power to compensate, which only exacerbates the problem and may very well lead to an overheated processor—and a system shutdown.

There are two types of leakage currents—gate oxide tunnelling current and sub-threshold leakage. As the die shrinks, the transistors get smaller alongside, and the insulation on these transistors also needs to get thinner. The thinning insulation results in leakage in the form of gate oxide tunnelling current. Sub-threshold leakage is the leakage current flowing through a transistor when it is supposed to be turned off.

The way a transistor works, is similar to a gate. It stops the flow of current from one side of the gate to the other side. The gate is closed until the voltage across the transistor is less than a certain 'threshold'. As the voltage crosses this threshold, the gate opens up and lets the current through. Thus, the transistor is either ON or OFF depending upon this threshold voltage. As long as the threshold is not reached, the transistor should remain OFF, and no current should flow through the transistor.